FOLIC ACID

Neural tube defect prevention

Check B12 first in anaemia

CHEMICAL NAME (2S)-2-[[4-[(2-amino-4-oxo-1H-pteridin-6-yl)methylamino]benzoyl]amino]pentanedioic acid (pteroylglutamic acid, vitamin B9)

CLASS Water-soluble B vitamin (folate)

TRADE NAMES Folic acid 400 mcg / 5 mg · Folicare · Lexpec (syrup) · Folvite

THERAPEUTIC USES / INDICATIONS

- Prevention of neural tube defects; before conception and until 12 weeks of pregnancy.
- Folate-deficiency megaloblastic anaemia.
- Haemolytic anaemias (sickle cell, thalassaemia), dialysis, malabsorption.
- Reduce methotrexate toxicity (weekly); with antiepileptics in pregnancy.

ADMINISTRATION

- **PO** once daily (any time).
- Methotrexate patients: once weekly on a different day from methotrexate.

DOSAGES

- **NTD prevention (low risk):** 400 mcg daily from ≥1 month before conception until 12 weeks.
- **High risk** (previous or family history of NTD, type 1/2 diabetes, sickle cell or thalassaemia, antiepileptic or HIV medicines): 5 mg daily.
- **Megaloblastic anaemia:** 5 mg daily for 4 months (exclude/treat B12 deficiency first).
- **Methotrexate:** 5 mg once weekly.

PHARMACOKINETICS

- Absorbed in proximal small intestine; converted to active 5-methyltetrahydrofolate.
- Stored in liver; excreted in urine and bile.

PHARMACODYNAMICS

- Essential cofactor for **DNA synthesis** (thymidylate and purines) and amino-acid metabolism (homocysteine → methionine) → normal cell division, red-cell maturation, neural tube closure.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Rare: GI upset, bloating.
- Allergic reactions (rare).

ADVERSE EFFECTS (SERIOUS)

- Masks B12 deficiency; corrects anaemia while subacute combined degeneration of the cord progresses.
- High doses may reduce antiepileptic levels.

CONTRAINDICATIONS & CAUTIONS

- Untreated B12 deficiency (unless with B12).
- Malignancy (folate-dependent tumours — caution).

KEY INTERACTIONS

- **Phenytoin, phenobarbital, primidone:** ↓ levels (folic acid) and folate deficiency (antiepileptics).
- **Methotrexate, trimethoprim, pyrimethamine:** antagonists of folate.
- **Sulfasalazine:** ↓ folate absorption.

NURSING MANAGEMENT

- Check B12 and folate before treating megaloblastic anaemia.
- Advise all women planning pregnancy about folic acid.
- Monitor FBC response (reticulocytosis within 1 week).

PATIENT EDUCATION

- Start folic acid before you try to conceive and continue to 12 weeks.
- Eat folate-rich foods (green leafy vegetables, legumes, fortified cereals).
- Take on a different day from methotrexate.

KEY POINTS

- Vitamin B9 — DNA synthesis.
- Check B12 first in anaemia.
- 400 mcg (low risk) or 5 mg (high risk) for NTD prevention.
- Weekly with methotrexate.

SOURCES CHECKED [934] NICE NG247

Full references p. 432–457